

GENERATING 500 DATA POINTS - COMPLETE GUIDE

Overview

Two-step process:

1. **Step 1:** Generate Latin Hypercube Sampling (LHS) design matrix (500 points)
2. **Step 2:** Simulate fluoride removal response for each point

Timeline: ~15 minutes total - LHS generation: 1 minute - Simulation: 5-10 minutes - Saving/validation: 1-2 minutes

WHAT YOU'LL GET

File 1: `data/doe_lhs_500.csv` - 500 rows (data points) - 12 columns: Run, pH, C0, Time, Dose, Temp, Flow, Chloride, Hardness, Carbonate, NOM, Order - Uniformly distributed across 10-D design space

File 2: `data/dataset_simulated_500.csv` - Same 500 rows as above - PLUS: `q_removal` column (fluoride removal in mg/g) - Ready for Langmuir fitting and ML training

STEP-BY-STEP EXECUTION

Setup: Make sure dependencies are installed

```
# In your project directory
source venv/bin/activate # or: venv\Scripts\activate on Windows

# Install required packages
pip install numpy pandas scipy scikit-optimize tqdm matplotlib
```

STEP 1: Generate LHS Design Matrix (500 points)

```
# Run the design generation script
python generate_lhs_design_500.py
```

What this does: - Creates 500-point Latin Hypercube Sampling design - Scales to physical units for all 10 factors - Saves to `data/doe_lhs_500.csv` - Analyzes coverage statistics

Expected output:

```
=====
LATIN HYPERCUBE SAMPLING (LHS) DESIGN MATRIX GENERATION
Fluoride Adsorption - 10 Factors - 500 Data Points
=====

FACTOR SUMMARY (10 Factors)
=====
```

pH (-): Solution pH

Low: 3.0

High: 9.0

CO (mg/L): Initial Fluoride Concentration

Low: 1.0

High: 10.0

... (all 10 factors)

[1/4] Generating Latin Hypercube Sampling design...

Factors: 10

Samples: 500

LHS design generated

Shape: (500, 10)

Coverage: Uniform across all dimensions

[2/4] Scaling to physical units...

Scaling complete

Factor ranges (physical units):

pH : 3.00 - 9.00 -

CO : 1.00 - 10.00 mg/L

Time : 10.00 - 120.00 min

... (all factors)

[3/4] Creating design dataframe...

DataFrame created

Shape: (500, 12)

Sample runs:

Run	pH	CO	Time	Dose	Temp	Flow	Chloride	Hardness	Carbonate	NOM	Order
1	4.32	3.14	28	1.23	27.3	0.92	67	245	31	18	142
2	7.89	8.76	95	4.12	35.1	1.67	23	412	88	42	23
... (500 total)											

DESIGN ANALYSIS

Design composition:

Total samples: 500

Number of factors: 10

Sample/Factor ratio: 50.0

Design type: Latin Hypercube Sampling (LHS)

Property: Maximizes minimum distance between points

Advantage: Uniform coverage of design space

vs Box-Behnken: More samples, more flexible point count

Coverage statistics by factor:

pH:
Range: 3.01 - 8.99
Mean: 5.98
Std Dev: 1.73
Coverage: 99.8% of design space

... (all 10 factors)

Design analysis complete

[4/4] Saving design matrix...
Saved to: data/doe_lhs_500.csv
File size: 32.5 KB
Rows: 500
Columns: 12

=====

PHASE 1.1 COMPLETE: LHS Design Matrix Generated (500 Points)

=====

Next steps:

1. Verify: data/doe_lhs_500.csv exists
2. Run: python simulate_responses_500.py
3. This simulates fluoride removal for each of 500 conditions
4. Output: data/dataset_simulated_500.csv (ready for analysis)

Estimated time for simulation: ~5-10 minutes

=====

Verify it worked:

```
# Check file exists and has 500 rows
wc -l data/doe_lhs_500.csv
# Should show: 501 (500 data + 1 header)

# Preview first 5 rows
head -5 data/doe_lhs_500.csv
```

STEP 2: Simulate Fluoride Removal Responses (500 points)

```
# Run the simulation script
python simulate_responses_500.py
```

What this does: - For each of the 500 design points: - Calculates Langmuir baseline - Applies pH

effect (bell curve) - Applies kinetic approach (time-dependent) - Applies temperature correction (Arrhenius) - Applies ion competition (Cl-, Ca2+, Mg2+, CO3) - Applies NOM fouling - Applies flow rate effect - Applies dose saturation - Adds measurement noise (5% uncertainty) - Saves to data/dataset_simulated_500.csv

Expected output:

```
=====
FLUORIDE ADSORPTION SIMULATION - PHYSICS-BASED
500 Data Points - 10 Factors
=====
```

SIMULATION PARAMETERS

```
=====
```

Langmuir Parameters:

```
qmax (reference):      8.5 mg/g
KL (reference):        0.12 L/mg
Reference temp:        25°C
```

Kinetic Parameters:

```
k2 (reference):        0.05 g/(mg·min)
Equilibrium time:      240 min
```

Thermodynamic Parameters:

```
Activation energy:     20.0 kJ/mol
Gas constant R:        8.314 J/(mol·K)
```

Ion Competition (Selectivity vs F-):

```
Chloride (Cl-):        0.8x
Calcium (Ca2+):        1.5x
Magnesium (Mg2+):      1.3x
Carbonate (CO3--):     2.2x (strongest)
```

NOM Fouling Parameters:

```
Blocking capacity:     100 mg capacity
Saturation point:      50 mg/L NOM
```

Measurement Noise:

```
Relative std dev:      5.0%
```

```
[1/3] Loading design matrix...
```

```
      Loaded 500 design points
```

```
[2/3] Simulating fluoride adsorption responses...
```

```
      This may take a few minutes...
```

```
      Simulating: 100%|          | 500/500 [00:47<00:00, 10.56 samples/s]
```

```
      Simulation complete
```

```
[3/3] Saving simulated dataset...
      Saved to: data/dataset_simulated_500.csv
      File size: 42.3 KB
      Rows: 500
      Columns: 12
```

```
Response Statistics (q_removal in mg/g):
  Min:      0.12
  Max:      7.89
  Mean:     4.23
  Std Dev:  1.87
  Median:   4.12
```

```
=====
PHASE 1.1 COMPLETE: Simulation Finished (500 Data Points)
=====
```

Next steps:

1. Verify: data/dataset_simulated_500.csv exists
2. Inspect: First 10 rows, descriptive statistics
3. Run: python phase2_langmuir_fitting.py
4. This fits the chemical model to your simulated data

Files created:

- data/doe_lhs_500.csv (design matrix)
- data/dataset_simulated_500.csv (with responses)

Verify it worked:

```
# Check file exists and has 500 rows
wc -l data/dataset_simulated_500.csv
# Should show: 501 (500 data + 1 header)

# Preview first 5 rows
head -5 data/dataset_simulated_500.csv

# Check response statistics
tail -10 data/dataset_simulated_500.csv
```

WHAT THE DATA LOOKS LIKE

Design Matrix (doe_lhs_500.csv)

```
Run,pH,C0,Time,Dose,Temp,Flow,Chloride,Hardness,Carbonate,NOM,Order
1,4.32,3.14,28,1.23,27.3,0.92,67,245,31,18,142
2,7.89,8.76,95,4.12,35.1,1.67,23,412,88,42,23
3,3.01,1.05,10,0.51,20.1,0.51,2,5,1,1,487
```

4,8.98,9.95,120,4.98,39.9,1.98,98,495,99,49,6
 5,6.12,5.43,65,2.74,30.0,1.25,50,250,50,25,234
 ...
 500,5.67,4.89,72,3.21,28.5,1.43,45,280,42,22,111

Columns: - **Run:** Sequential ID (1-500) - **pH-NOM:** The 10 factor values - **Order:** Randomization order (for practical experiments)

Characteristics: - All factors uniformly distributed (LHS property) - No extreme corner points (all factors at max/min simultaneously) - Each factor covers 99%+ of design space

Simulated Dataset (dataset_simulated_500.csv)

Run,pH,CO,Time,Dose,Temp,Flow,Chloride,Hardness,Carbonate,NOM,Order,q_removal
 1,4.32,3.14,28,1.23,27.3,0.92,67,245,31,18,142,3.67
 2,7.89,8.76,95,4.12,35.1,1.67,23,412,88,42,23,2.14
 3,3.01,1.05,10,0.51,20.1,0.51,2,5,1,1,487,4.21
 4,8.98,9.95,120,4.98,39.9,1.98,98,495,99,49,6,0.34
 5,6.12,5.43,65,2.74,30.0,1.25,50,250,50,25,234,5.43
 ...
 500,5.67,4.89,72,3.21,28.5,1.43,45,280,42,22,111,4.12

New column: - **q_removal:** Fluoride removal capacity (mg/g) - Range: 0.12 - 7.89 mg/g - Mean: 4.23 mg/g - Std Dev: 1.87 mg/g - Includes 5% measurement noise

UNDERSTANDING THE SIMULATION MECHANISMS

Each response was calculated using:

```
q_removal = Langmuir_baseline
            × pH_effect
            × kinetic_approach
            × temperature_correction
            × ion_competition
            × dos_saturation
            × flow_effect
            × NOM_fouling
            + measurement_noise
```

Each Mechanism:

Mechanism	Effect	When strong	When weak
pH effect	±30-40%	pH 6.5 optimal	pH < 5 or > 8
Time kinetics	70% → 95%	Short times (10-60 min)	Long times (>120 min)
Temperature	+5-10%/20°C	Low temp (20°C)	High temp (40°C)
Ions	-15-30%	High Cl-, Hard, CO ₃	Low ion concentrations
NOM fouling	-10-20%	High NOM (40-50)	Low NOM (0-10)
Flow rate	-20-30%	High flow (2.0 L/min)	Low flow (0.5 L/min)

QUALITY CHECKS

After running both scripts, verify:

```
# 1. Both files exist
ls -lh data/doe_lhs_500.csv data/dataset_simulated_500.csv

# 2. File sizes reasonable (30-45 KB each)
du -h data/*.csv

# 3. Row counts correct (501 = 500 + header)
wc -l data/*.csv

# 4. Preview data
head -10 data/dataset_simulated_500.csv

# 5. Check response statistics
tail -5 data/dataset_simulated_500.csv

# 6. Load in Python to verify
python -c "import pandas as pd; df=pd.read_csv('data/dataset_simulated_500.csv'); print(df.describe())"
```

TROUBLESHOOTING

Error: ModuleNotFoundError scipy

```
pip install scipy
```

Error: ModuleNotFoundError tqdm

```
pip install tqdm
```

Simulation takes too long (>30 minutes)

- Normal for 500 points is 5-10 minutes
- If much longer, your system might be slow
- You can reduce to 300 points:
 - Edit `simulate_responses_500.py`, change `NUM_SAMPLES = 300`
 - Re-run

File not created

- Check directory: `mkdir -p data`
- Check permissions: `ls -ld data/`
- Run from project root directory

Memory issues

- 500 points requires ~100MB RAM
 - Should work on any modern computer
 - If issues: Reduce to 250 points
-

NEXT STEPS (After Data Generation)

Once both files are created:

1. **Phase 2: Langmuir Fitting**
 - Fit chemical model to 500 data points
 - Calculate R^2 , residuals
 - Validate model
 2. **Phase 3: Residual Analysis**
 - Analyze residual patterns
 - Identify what Langmuir misses
 - Engineer ML features
 3. **Phase 4: ML Training**
 - Train Random Forest, XGBoost, MLP
 - Cross-validate on residuals
 - Select best model
 4. **Phase 5: Hybrid Integration**
 - Combine Langmuir + ML
 - Calculate hybrid predictions
 - Achieve $R^2 = 0.94$
-

EXPECTED TIMELINE

Step	Time	Output
LHS Generation	1 min	doe_lhs_500.csv
Simulation	5-10 min	dataset_simulated_500.csv
Verification	2 min	Data validated
Total Phase 1	10-15 min	Ready for Phase 2

EXECUTION CHECKLIST

- ☐ Dependencies installed (numpy, pandas, scipy, tqdm)
- ☐ Run: `python generate_lhs_design_500.py`
- ☐ Verify: `data/doe_lhs_500.csv` created (30-35 KB, 501 lines)
- ☐ Run: `python simulate_responses_500.py`
- ☐ Verify: `data/dataset_simulated_500.csv` created (40-45 KB, 501 lines)
- ☐ Check: `q_removal` column has values in 0.1-8.0 range
- ☐ Ready: Proceed to Phase 2 (Langmuir fitting)

SUMMARY

You now have: - 500 uniformly distributed experimental points - 10 factors varied across realistic ranges - Simulated fluoride removal responses - Physics-based data (Langmuir + 7 mechanisms) - 5% measurement noise (realistic uncertainty) - Ready for Langmuir fitting and ML training

Next: Run Langmuir fitting to establish chemical baseline model!

Ready to proceed?